

Chapter 7

Bayes in Practice; Random Variables

Lecture 7 | Devore 9e: 2.4–3.1

Recall from last time:

$$\underbrace{\mathbb{P}(A_i | B)}_{\text{posterior}} = \frac{\overbrace{\mathbb{P}(B | A_i)}^{\text{likelihood}} \overbrace{\mathbb{P}(A_i)}^{\text{prior}}}{\sum_j \mathbb{P}(B | A_j) \mathbb{P}(A_j)}.$$

7.1 Disease testing, finished

Example 7.1 (Rare disease). A disease affects 1% of the population:

$$A = \{\text{person has the disease}\}, \quad \mathbb{P}(A) = 0.01.$$

The test satisfies $\mathbb{P}(B | A) = 0.99$ (sensitivity) and $\mathbb{P}(B^c | A^c) = 0.95$ (specificity), where $B = \{\text{test is positive}\}$; hence $\mathbb{P}(B | A^c) = 0.05$. A patient tests positive. What is $\mathbb{P}(A | B)$?

$$\begin{aligned} \mathbb{P}(A | B) &= \frac{\mathbb{P}(A)\mathbb{P}(B | A)}{\mathbb{P}(A)\mathbb{P}(B | A) + \mathbb{P}(A^c)\mathbb{P}(B | A^c)} \\ &= \frac{(0.01)(0.99)}{(0.01)(0.99) + (0.99)(0.05)} = \frac{0.0099}{0.0594} = \boxed{\frac{1}{6} \approx 16.7\%} \end{aligned}$$

Remark 7.2 (The base-rate trap). The test is right 99% of the time on sick people, yet a positive result means only a 1-in-6 chance of being sick. Here is the back-of-envelope version, and it is worth teaching this way:

$$\text{positive result} \implies \begin{array}{ll} \text{either} & \text{the person is sick} \quad (1\% \text{ of people}) \\ \text{or} & \text{the test is wrong} \quad (5\% \text{ of the other } 99\%) \end{array}$$

and $\frac{1}{1+5} = \frac{1}{6}$. Because the disease is rare, the (small) false-positive *rate* acts on a (huge) healthy population, and the false positives outnumber the true positives five to one.

Now check the negative result. With $\mathbb{P}(B^c | A) = 0.01$,

$$\mathbb{P}(A^c | B^c) = \frac{(0.99)(0.95)}{(0.99)(0.95) + (0.01)(0.01)} = \frac{0.9405}{0.9406} \approx 99.99\%.$$

So for a *rare* disease:

positive $\Rightarrow$ has the disease $\times$ (only 16.7%)
 negative $\Rightarrow$ does not $\checkmark$ (99.99%)

Example 7.3 (Common disease — the pattern reverses). Same test, but now $\mathbb{P}(A) = 0.99$. Then

$$\mathbb{P}(A | B) = \frac{(0.99)(0.99)}{(0.99)(0.99) + (0.01)(0.05)} = \frac{0.9801}{0.9806} \approx 99.9\%,$$

$$\mathbb{P}(A^c | B^c) = \frac{(0.01)(0.95)}{(0.01)(0.95) + (0.99)(0.01)} = \frac{0.0095}{0.0194} \approx 49\%.$$

Now a positive result is conclusive and a *negative* result is a coin flip.

Remark 7.4. The test never changed. What changed is $\mathbb{P}(A)$. This is the sentence to remember: $\mathbb{P}(A | B)$ is not a property of the test; it is a property of the test **and** the population.

Remark 7.5 (Careful: the shortcut does not survive the switch). The “ $\frac{1}{1+5}$ ” reasoning above works for the *rare* disease because $\mathbb{P}(A) = 1\%$ is small enough that the healthy group is essentially the whole population, so its false-positive mass is just 5%. Running the same shortcut on the common disease gives

$$\frac{99}{99+5} = \frac{99}{104} \approx 95\%,$$

which is **wrong** — it charges the full 5% false-positive rate to a group that is only 1% of the population. The correct weighting is $\mathbb{P}(A^c)\mathbb{P}(B | A^c) = (0.01)(0.05) = 0.0005$, giving 99.9% as computed above. When the base rate is not small, do the arithmetic in full.

7.1.1 Testing twice

Example 7.6. Rare disease again, $\mathbb{P}(A) = 0.01$. Test the patient twice; let B_1, B_2 be the two positive results, and assume the two tests are *conditionally independent* given the true status:

$$\mathbb{P}(B_1 \cap B_2 | A) = \mathbb{P}(B_1 | A)\mathbb{P}(B_2 | A), \quad \mathbb{P}(B_1 \cap B_2 | A^c) = \mathbb{P}(B_1 | A^c)\mathbb{P}(B_2 | A^c).$$

Then

$$\begin{aligned} \mathbb{P}(A | B_1 \cap B_2) &= \frac{\mathbb{P}(B_1 | A)\mathbb{P}(B_2 | A)\mathbb{P}(A)}{\mathbb{P}(B_1 | A)\mathbb{P}(B_2 | A)\mathbb{P}(A) + \mathbb{P}(B_1 | A^c)\mathbb{P}(B_2 | A^c)\mathbb{P}(A^c)} \\ &= \frac{(0.99)^2(0.01)}{(0.99)^2(0.01) + (0.05)^2(0.99)}. \end{aligned}$$

Numerically $\frac{0.009801}{0.009801 + 0.002475} \approx \boxed{80\%}$.

One positive: 16.7%. Two positives: 80%. This is why screening programs re-test rather than trusting a single result.

7.1.2 Bayes in odds form

There is a way of writing Bayes’ theorem that turns “testing twice” into mental arithmetic. The **odds** of an event are $O(A) = \mathbb{P}(A)/\mathbb{P}(A^c)$. Write Bayes’ theorem once for A and once for A^c , given the same evidence B , and divide: the common denominator $\mathbb{P}(B)$ cancels.

Theorem 7.7 (Bayes' theorem, odds form).

$$\underbrace{O(A | B)}_{\text{posterior odds}} = \underbrace{O(A)}_{\text{prior odds}} \times \underbrace{\frac{\mathbb{P}(B | A)}{\mathbb{P}(B | A^c)}}_{\text{likelihood ratio}}.$$

Each piece of evidence *multiplies* the odds by its likelihood ratio, and conditionally independent pieces of evidence simply multiply one after another. For the rare disease: the prior odds are 1 : 99 and the likelihood ratio is $0.99/0.05 = 19.8$, so after one positive the odds are $19.8 : 99 = 1 : 5$, i.e. $\mathbb{P}(A | B) = \frac{1}{6}$; after a second positive they are $1 : 5 \times 19.8 \approx 4 : 1$, i.e. about 80%. Both answers from above, with no denominators to compute.

Example 7.8 (Two signals). A trader thinks a stock is mispriced with probability 0.2. Two signals, conditionally independent given the truth, each fire with probability 0.6 if the stock is mispriced and 0.2 if it is not. Both fire. What is the probability that the stock is mispriced?

The prior odds are $0.2 : 0.8 = 1 : 4$ and each signal has likelihood ratio $0.6/0.2 = 3$, so the posterior odds are $\frac{1}{4} \times 3 \times 3 = \frac{9}{4}$ and

$$\mathbb{P}(\text{mispriced} | \text{both fire}) = \frac{9}{9 + 4} = \boxed{\frac{9}{13}} \approx 0.69.$$

Directly from Bayes:
$$\frac{0.2(0.6)^2}{0.2(0.6)^2 + 0.8(0.2)^2} = \frac{0.072}{0.104} = \frac{9}{13}.$$

Example 7.9 (An unreliable witness). Alice and Bob each tell the truth with probability $\frac{1}{3}$, independently. Alice makes a statement, and Bob then says, “Alice told the truth.” What is the probability that Alice really did tell the truth?

Let T be the event that Alice told the truth, so the prior odds are 1 : 2. Bob's claim happens if Alice was truthful and Bob is too, or if Alice lied and Bob lies about it:

$$\mathbb{P}(\text{claim} | T) = \frac{1}{3}, \quad \mathbb{P}(\text{claim} | T^c) = \frac{2}{3}.$$

The likelihood ratio is $\frac{1}{2}$, the posterior odds are 1 : 4, and $\mathbb{P}(T | \text{claim}) = \boxed{\frac{1}{5}}.$

Remark 7.10. Bob's endorsement made Alice *less* credible, from $\frac{1}{3}$ down to $\frac{1}{5}$. Evidence raises the odds only when its likelihood ratio exceeds 1; a witness who is wrong more often than right is evidence for the opposite of what they say.

7.2 Random variables

We rarely care about the outcome ω itself; we care about some *number* attached to it. That number is a random variable.

Example 7.11. $S = \{\text{heart, diamond, spade, club}\}$ and

$$X(\text{heart}) = X(\text{diamond}) = 0, \quad X(\text{spade}) = X(\text{club}) = 1.$$

The new “sample space” — the set of values — is $\hat{S} = \{0, 1\}$.

Example 7.12. A gambler plays n games: $S = \{W, L\}^n$, so $|S| = 2^n$. Let X be the number of wins. Then $X(WLL \cdots L) = 1$, $X(LWLL \cdots L) = 1$, and $\widehat{S} = \{0, 1, 2, \dots, n\}$, of size $n + 1$.

Note the compression: 2^n outcomes collapse to $n + 1$ values. That is the point — a random variable throws away the information we do not need.

Example 7.13 (Continuous). Throw a dart at the board $S = \{(x, y) : |x| \leq 1, |y| \leq 1\}$ and record the distance from the center:

$$X(x, y) = \sqrt{x^2 + y^2}, \quad \widehat{S} = [0, \sqrt{2}].$$

Definition 7.14 (Random variable). Let $(S, \mathcal{F}, \mathbb{P})$ be a probability space. A function $X : S \rightarrow \mathbb{R}$ is a **random variable** if for every $x \in \mathbb{R}$,

$$\{\omega \in S : X(\omega) \leq x\} \in \mathcal{F}.$$

Remark 7.15. Why the condition? Because $\mathbb{P}$ is only defined on $\mathcal{F}$. If the set $\{X \leq x\}$ were not an event, the symbol $\mathbb{P}(X \leq x)$ would be meaningless — and that symbol is the entire subject of the next lecture. When $\mathcal{F} = \mathcal{P}(S)$ (any finite or countable S) the condition is automatic, so in this course it is never an obstacle; it is there so the definition also works on $\mathbb{R}$.

Remark 7.16 (Notation). In lecture I often write ξ (xi) for a random variable, and so do several homework problems; the textbook and the exams write X . They mean the same thing. Uppercase is the random variable (a *function*); lowercase x is a particular value it might take (a *number*). Writing $\mathbb{P}(X \leq x)$ is meaningful; writing $\mathbb{P}(x \leq x)$ is not.